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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25VDE132 - VLSI DESIGN VERIFICATION

(For VDE)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Answer Five full questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

Module – I

	Marks	BT*	CO*	PO*
1. a) How do emerging trends in VLSI technology influence testing methodologies? Provide a detailed justification with relevant explanations.	10	2	1	3
b) Provide a comprehensive analysis of system testbench metrics and their significance in validating design functionality.	10	2	2	3
2. a) What factors contribute to the failure of fabricated semiconductor products, and how does testing ensure fault-free functionality?	10	2	1	3
b) Elaborate on formal model checking and formal equivalence checking, highlighting their methodologies, applications, and significance in hardware verification.	10	2	2	3
3. a) Discuss System-on-Chip (SoC) verification technologies in detail, supported by a clear and structured design flow diagram.	10	2	1	3
b) What are formal verification technologies, and how are they applied in design validation? Briefly describe the key types used in modern verification workflows	10	2	2	3

Module – II

4. a) For the circuit shown in Figure 4(a), what is the maximum frequency of operation? Are there any hold time violations for FF2? If yes, how to modify the circuit to avoid them?

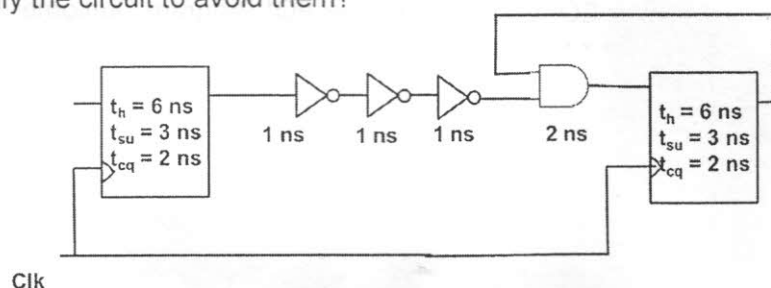


Figure 4a: Logic Diagram

b) How do IR drop and electromigration impact signal integrity in VLSI circuits? Justify with appropriate reasoning.	10	3	3	3
5. a) Define the following timing parameters in digital circuit design: (i) Setup Time, (ii) Hold Time, and (iii) Propagation Delay.	10	2	4	3
	6	2	3	3

- b) For the state machine shown in Figure 5(b), find the maximum frequency of operation. Given $t_{su} = 1.5$ ns and maximum clock skew is 0.5 ns.

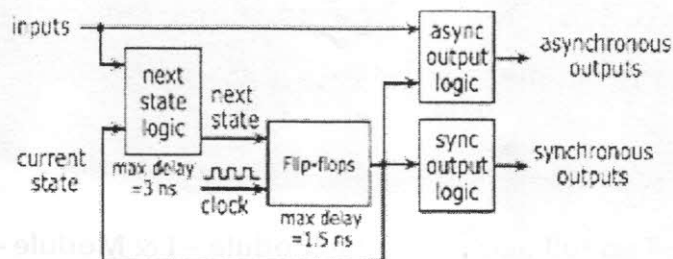


Figure 5b: State Machine

- c) What is crosstalk in VLSI circuits? Discuss its causes and explain how it affects signal integrity.

4	3	3	3
10	2	4	3

6. a) Explain the concepts of false path and latch borrowing in digital timing analysis, using appropriate examples to illustrate their significance
b) Explain PVT variations in detail, and discuss the concept of analysis corners in VLSI design verification.

10	2	3	3
10	2	4	3

Module – III

7. a) Briefly explain characterization testing and production testing in VLSI chip validation. Provide a comparative overview highlighting their objectives, methodologies, and applications.
b) Explain the concept of DC parametric testing in detail, highlighting its objectives, key parameters measured, and its role in ensuring semiconductor device reliability.
8. a) What is the Built-In Self-Test (BIST) process in VLSI design? Explain the architecture of BIST in detail with a well-labeled diagram
b) Explain the architecture and functionality of the Advantest Model T6682 Automatic Test Equipment (ATE) with the help of a well-labeled block diagram

10	2	5	3
10	2	5	3
10	2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
